

RESEARCH ARTICLE



The risk of acute disseminated encephalomyelitis (ADEM) following covid-19 vaccination in England: A self-controlled case-series analysis

Julia Stowe, Jamie Lopez-Bernal, and Nick Andrews

Immunisation Division, UK Health Security Agency, London, UK

ABSTRACT

Acute disseminated encephalomyelitis (ADEM) has been identified as an Adverse Event of Special Interest in the COVID-19 vaccine programme due to its long-standing temporal association with a wide range of other vaccines. Case reports of ADEM shortly following COVID-19 vaccination have now been documented. There were 217 ADEM admissions in 215 individuals in the period 8th December 2020 to 31st March 2023. An increased risk of ADEM following the first dose of ChAdOx1 vaccine was observed (relative incidence (RI) = 3.13, 95% Confidence Interval (CI) [1.56–6.25]) with a vaccine attributable risk of 0.39 per million doses. When doses 1 and 2 were combined this increased risk remained just significant (1.96 [95%CI 1.01–3.82]). No significant increased risk was observed with any other vaccine or dose. This small, elevated risk after the first dose of ChAdOx1-S vaccine demonstrates how large national electronic datasets can be used to identify very rare risks and provides reassurance that any risk of ADEM following the ChAdOx1-S COVID-19 vaccination is extremely small. Given the rarity of this risk, further studies in settings with access to data on large populations should be carried out to verify these findings.

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Introduction

Acute disseminated encephalomyelitis (ADEM) is a serious, but rare, disease which causes inflammation of the central nervous system damaging the brain's myelin leading to neurological symptoms such as seizures, limb paralysis and can be fatal.¹ It mostly occurs in children and young adults and typically presents with an acute onset after a bacterial or viral infection, including COVID-19.² Individual cases have also been reported temporally associated to a range of vaccines including the smallpox, rabies, human papilloma virus³ and influenza vaccines, but this temporal post-vaccine ADEM is rare and estimated to be only 1 to 2 - per million vaccinated individuals.⁴

Due to these long-standing documented occurrences after vaccination, ADEM has been included as an adverse event of special interest (AESI) following the COVID-19 vaccine roll-out, indicating that regulators and public health authorities should monitor cases and investigate should a signal occur.⁵ Following the global administration of the COVID-19 vaccine, case reports of ADEM following COVID-19 vaccines were reported in adults. A systematic review reported 67% of cases occurred after the ChAdOx1-S vaccine and 21% after the BNT162b2 vaccine mainly after the first dose with a median interval from vaccine of 26 days.⁶

In the United Kingdom two mRNA vaccines (mRNA-1273 vaccine and BNT162b2 vaccine) and the adenovirus-vectored vaccine (ChAdOx1-S) were predominantly used in the primary course with the booster doses mainly restricted to the mRNA vaccines. Uptake of the COVID-19 vaccine was high in those 50 years of age and over with over 90% receiving a dose

of vaccine by June 2021. In those 49 years of age and younger the uptake was lower with under 60% receiving a dose of vaccine by June 2021.⁷

Using the self-controlled case-series method (SCCS) with a national linked dataset, we estimate the risk of ADEM 0 to 42 days after COVID-19 vaccination stratified by dose and vaccine type for the primary course, ancestral strain monovalent boosters, and the autumn 2022 BA.1/2 bivalent booster.

Materials and methods

Study population

The study population consisted of individuals resident in England aged 5 years and older with a hospital admission for ADEM in the period 8th December 2020 to 31st March 2023.

Data sources

The Secondary Uses Service (SUS) database,⁸ which provides ICD-10 coded diagnoses for individuals discharged from all National Health Service hospitals in England, was used for case ascertainment. Finished Consultant Admissions for a ADEM (ICD-10 G040) in the first three diagnosis fields with at least 365 days since a prior hospitalization were identified.

Immunisation data were obtained from the National Immunisation Management System (NIMS) which records all COVID-19 vaccinations given in England. Primary course vaccination (dose 1 or 2) with either the ChAdOx1-S or mRNA (BNT162b2 or mRNA-1273) vaccines and booster vaccination

(≥3 doses) with ancestral strain monovalent or BA.1/2 bivalent mRNA vaccines were extracted. Date of death and clinical risk group flags, such as clinically vulnerable⁹ or severe immunosuppression, were extracted from the NIMS denominator file which contains the information on all individuals resident in England.

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Data linkage

SUS admission records were linked to the NIMS data set using NHS number. The extracted SUS admissions with the outcomes of interest that did not link with a NIMS record were excluded from the analysis; these comprised 0.3% of the extracted SUS admissions.

Statistical methods

The relative incidence (RI) of ADEM after COVID-19 vaccination was assessed using the SCCS method.¹⁰ Admissions to hospital with ADEM in the 0–42 days after the first and second ChAdOx1-S or mRNA (BNT162b2 or mRNA-1273) vaccines were assessed. The risk was also assessed in the 0–42 days after dose 3 or 4 of the mRNA vaccines and the mRNA bivalent vaccines. These risks were relative to the baseline incidence of ADEM outside these 0–42 day periods and within the study period. To account for individuals not getting vaccinated due to being in hospital for ADEM the period 1–14 days prior to vaccine was removed from the baseline except prior to dose 2 to avoid overlap with the 42 days risk period post dose 1. A period adjustment in 4-weekly intervals was included in

the model. A sensitivity analysis was carried out combining all ChAdOx1-S doses, all mRNA monovalent and restricting to ADEM ICD-10 G040 in the primary (first) diagnosis field. We also report the number of deaths within 90 days of the ADEM admission by vaccination status and perform a sensitivity analysis removing 20 cases who died to assess potential bias with the SCCS method when outcomes can lead to death.

Attributable risk estimates

Attributable cases in the risk intervals with elevated RIs were estimated from the attributable fraction $AF = (RI - 1)/RI$ multiplied by the number of cases in that interval. Attributable risk was then calculated from the attributable cases divided by the number of doses administered in the study period.

Results

There was a total of 217 hospital admissions in England with the ICD-10 code for ADEM in the first 3 diagnosis fields with 173 (79.7%) in the primary field and 2 individuals having a subsequent admission for ADEM in the study period. The numbers were consistent over time with a small drop in quarter 1, 2023 that may reflect some episodes not available in SUS at the time of analysis (Table 1). Of the 66 who never received a vaccine dose in the study period, 34 (51.5%) were between 5 and 9 years of age, with 19 (28.8%) aged between 10 and 19, 7 (10.6%) aged between 20 and 39, 6 (9.1%) aged 40–64 with no unvaccinated ADEM admissions aged over 65. Of those vaccinated the proportion that were clinically vulnerable or severely immunosuppressed was much higher in those who received ChAdOx1-S as a first dose (ChAdOx1-S vaccine 26/73 [37.0%] vs mRNA vaccine 8/78 [10.3%]).

Table 1. Description of the 215 individuals with an ADEM diagnosis from SUS in the period 8th December 2020 to 31st March 2023.

Factor	Level	Frequency of Identified ADEMs (n = 215)	% of Identified ADEMs
Period (year-quarter)	2020–4 (from Dec 8th)	6	2.8%
	2021–1	30	14.0%
	2021–2	27	12.6%
	2021–3	20	9.3%
	2021–4	24	11.2%
	2022–1	23	10.7%
	2022–2	21	9.8%
	2022–3	23	10.7%
	2022–4	26	12.1%
	2023–1	15	7.0%
Age-years	5 to 9	39	18.1%
	10 to 19	45	20.9%
	20 to 39	47	21.9%
	40 to 64	63	29.3%
	65+	21	9.8%
Sex	Male	100	46.5%
	Female	115	53.5%
Clinically Vulnerable or Immunosuppressed	No	176	81.9%
	Yes	41	19.1%
Ethnicity	White	157	73.0%
	Asian or Asian British	29	13.5%
	Black African, Caribbean or Black British	8	3.7%
	Mixed or Multiple Ethnic Groups	6	2.8%
	Not Known or Other	15	7.0%
Ever Vaccinated	No	66	30.7%
	Yes (and first dose ChAdOx1-S)	73	34.0%
	Yes (and first dose mRNA)	76	35.3%

Table 2. Relative incidence of ADEM by vaccine type and dose.

Vaccine Type	Description	Risk Window	Person Days	Events	Relative Incidence (95% Confidence Interval)
Control time	Baseline	none	160,145	184	Base
Pre-vaccine 14 day windows*	Removed	−14 to −1	3717	1	n/a
ChAdOx1-S	Vaccine dose 1	0–42 days	3136	12	3.13 (1.56–6.25)
	Vaccine dose 2	0–42 days	2494	1	0.36 (0.05–2.68)
mRNA	Vaccine dose 1	0–42 days	3196	3	0.78 (0.24–2.50)
	Vaccine dose 2	0–42 days	2838	6	1.77 (0.76–4.13)
	Vaccine dose 3 or 4	0–42 days	4515	7	1.26 (0.56–2.85)
	Bivalent vaccine BA1/2	0–42 days	1419	3	1.98 (0.58–6.80)

*Pre-vaccination period 1–14 days prior to vaccine was removed to account for individuals not getting vaccinated due to being in hospital for ADEM.

There was evidence of an increased risk of ADEM after the first dose of ChAdOx1-S vaccine with 12 events in the 42 days following vaccination (RI = 3.13, 95% CI [1.56–6.25]) (Table 2). No other vaccine or dose gave a significant increased risk. When doses 1 and 2 were combined and doses 1 to 4 were combined for the mRNA vaccine the increased risk was just significant (1.96 [95%CI 1.01–3.82]) for ChAdOx1-S (Table S1). Assessment of the number of events in four 6-week periods after the first dose of ChAdOx1-S showed that following the 12 events during the first 6 weeks there were 4,3,5 events in the next three six-week intervals. From the 217 admissions in the study there was a total of 12 deaths (5.5%) within 90 days of admission with only two following an episode that occurred in a vaccine risk window from a total of 32 admissions in the risk window (6.3%) (one in a case occurring after a first mRNA dose and one after a second dose).

When looking only at the 173 episodes where ADEM was in the first diagnostic field the results were similar (Tables S3,4) although the mRNA dose 2 estimate was close to significance (RI 2.38, 95% CI [0.99–5.65]), with a total of 6 events in the risk window of which one died within 90 days of admission. Removal of 20 cases who died in the study period gave similar results (Table S4).

The total number of ChAdOx1-S first doses given in the study period was 20.75 million. Of the 11 cases in the risk period an estimated $12 \times 2.13/3.13 = 8.2$ are vaccine attributable giving an attributable risk of 0.39 per million doses. If the mRNA dose 2 RI of 2.38 was a true risk, then this translates to 3.5 cases from a total of 22.9 million doses which is a risk of 0.15 per million doses. Similarly with 12.8 million doses of bivalent mRNA vaccine given any risk after this would also have to be very small.

Discussion

This is the first large national epidemiological study assessing the risk of ADEM after COVID-19 vaccination. Using established methods used to address many pertinent vaccine safety concerns^{11–14} we report evidence of a very small risk of ADEM following a first ChAdOx1-S vaccination of less than 1 in a million doses. There was no evidence of a risk of ADEM with a subsequent dose of ChAdOx1-S vaccine nor with any dose of mRNA vaccine.

The temporal association of neurological conditions after a vaccine have seldom been proven to be causally related once investigated using robust epidemiological methods.¹⁵ ADEM is similar to multiple sclerosis but ADEM is usually characterized by a single acute event where multiple sclerosis consists of relapsing progressive episodes.¹⁶ Although there are no

epidemiological studies assessing the risk of ADEM after vaccination, the risk of multiple sclerosis after hepatitis B vaccination has been investigated following passive reports during the 1994 mass hepatitis B vaccination campaign in France.¹⁷ A systematic review and meta-analysis by Mouchet et al. published in 2018 included 13 studies with a control group and found no evidence of an increased risk.¹⁸

The small increased risk reported in this study of ADEM following the first ChAdOx1-S vaccination is also seen in published case reports.⁶ In a literature review by Huynh et al. that included 54 cases of ADEM following the COVID-19 vaccine the mean age was 43.1 years with 61.1% being female which concurs with this study. This small risk is in contrast with the much higher risks of morbidity and mortality due to COVID-19 which were substantially reduced by vaccination, particularly in the period in 2021 after the vaccine was given when Delta was the predominant COVID-19 variant.^{19,20}

If this small risk after the ChAdOx1 vaccine demonstrated in this study is real, the potential mechanism is unclear. SARS CoV 2 itself has been associated with ADEM.² One theory is that immune stimulation could play a role in the pathogenesis via molecular mimicry or by nonspecific activation of an autoreactive T cell clone resulting in a transient autoimmune response against myelin.²¹ Short-term increased rates of other likely autoimmune neurological diseases have also been observed after the first dose of ChAdOx1 vaccine such as Guillain-Barré syndrome and Bell's palsy.^{22,23}

The SCCS method of analysis, being within-person deals with potential biases such as confounding by indication where the ChAdOx1-S vaccines were skewed toward the clinically vulnerable. Nevertheless, ADEM is a very rare diagnosis, so this analysis is based on small numbers with 12 events in the risk window and would benefit from validation in other studies. Relative incidence estimates were also slightly non-significantly elevated after mRNA second doses, but even if real this would be a risk in the region of 1 in 10 million. Main assumptions of the SCCS method are that events are independent and that they don't influence subsequent exposures.²⁴ In this study we used events with at least 365 days between them to insure they were independent admissions. ADEM is not a contraindication for vaccination so should not influence subsequent exposure other than the fact that if it causes death subsequent exposure cannot occur. We addressed this latter issue in our sensitivity analysis.

Although the diagnosis coding in this study was not validated, the sensitivity analysis using the ADEM ICD-10 diagnosis code only in the primary field resulted in a slightly higher estimate RI 4.08 (1.93–8.59) compared to using a diagnosis in

the first three diagnosis field RI 3.13 (1.56–6.25), although this was difference not significantly higher.

Overall, this study demonstrates the ability to detect very rare adverse outcomes by using large national electronic datasets and provides reassurance that if there is a risk of ADEM following the ChAdOx1-S COVID-19 vaccination then it is extremely small.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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The author(s) reported there is no funding associated with the work featured in this article.

Data availability statement

The raw study data are protected and are not available due to data privacy laws. This work is carried out under Regulation 3 of The Health Service (Control of Patient Information) (Secretary of State for Health, 2002)³ using patient identification information without individual patient consent. Data cannot be made publicly available for ethical and legal reasons, i.e. public availability would compromise patient confidentiality as data tables list single counts of individuals rather than aggregated data.

Governance

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